

Probabilistic Shaping and Forward Error Correction for Fiber-Optic Communication Systems

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(Invited Tutorial)

Abstract—This tutorial paper provides a foundation for integrating probabilistic shaping (PS) and forward error correction (FEC). A layered PS-FEC architecture consisting of a PS encoder and an FEC encoder is introduced, of which probabilistic amplitude shaping is a practical instance. Achievable PS encoding rates and achievable FEC decoding rates are derived using information-theoretic arguments. The developed tools are applied to the design and performance assessment of optical transponders based on measurements from optical transmission experiments.

Index Terms—Channel coding, coded modulation, distribution matching, optical fiber communication, probabilistic constellation shaping, probabilistic shaping.

I. INTRODUCTION

COMMUNICATION channels often have non-uniform capacity-achieving input distributions, which has been the main motivation for probabilistic shaping (PS), i.e., the development of practical transmission schemes that use non-uniform distributions at the input of the channel. Many different PS schemes have been proposed in literature, see, e.g., the literature review in [2, Section II].

An important milestone for making PS practical was the invention of probabilistic amplitude shaping (PAS) [2], which concatenates a shaping outer code called a distribution matcher [3] and a forward error correction (FEC) inner code. This “reverse concatenation” was originally proposed for constrained coding in [4], [5]. The PAS architecture has three properties that distinguishes it from other proposed PS schemes. First, it integrates shaping with existing FEC, second, it achieves the Shannon limit [6, Sec. 10.3], [7], third, it adapts its rate by changing the probability distribution, while leaving the FEC part unchanged.

Because of these three properties, PAS plays an important role in coherent optical transmission systems. It allows to increase

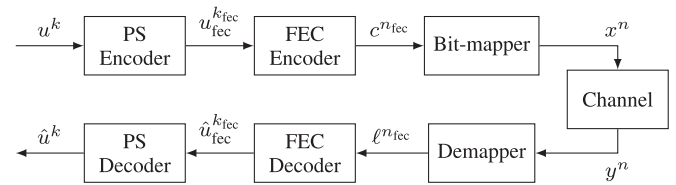


Fig. 1. Layered PS architecture.

the throughput of already deployed fiber infrastructure close to its theoretical limit and it enables an adjustment of the transmission rate with very fine granularity regarding given reach and link-level requirements. This overcomes the drawbacks of less flexible legacy systems.

The first demonstration of PAS in an optical transmission experiment [8] showed an increase in transmission reach of up to 40%. The presented results were obtained by a full implementation of PAS, including (at the transmitter) a distribution matching (DM) algorithm and an FEC encoder, and (at the receiver) an FEC decoder and an inverse DM algorithm. The processing of the recorded data was done offline. The results of [8] were confirmed, e.g., in [9]–[14]. Besides optical transmission experiments targeting different modulation formats, e.g., [15]–[19], PS for fiber-optic transmission was researched in mainly three directions. First, probability distributions tailored for transmission over the optical fiber were studied, e.g., [20]–[23]. Second, DM algorithms were proposed that can cope with the throughput requirements of optical transponders, e.g., [24]–[35]. Third, the design of FEC codes suitable for the use with non-uniform distributions was studied, e.g., [36]–[40].

The implementation of PS in optical transponders adds the shaping rate as an additional parameter that needs to be taken into account in system design and performance evaluation. For example, the overall overhead consists of the FEC overhead and the PS overhead. Mastering the additional PS parameter requires a substantial understanding of the interplay of PS and FEC.

As a tool for the joint design of PS and FEC, the central object in this tutorial is the following rate expression for the layered PS architecture in Fig. 1:

$$\left[\frac{\log_2 |\mathcal{S}|}{n} - \mathbb{U}(q, X, Y) \right]^+ \quad (1)$$

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where $[\cdot]^+ = \max(\cdot, 0)$. The first term in (1) contains the shaping set $\mathcal{S}$, which is determined by the shaping device, e.g., a DM. The value of $\log_2 |\mathcal{S}|/n$ accounts for the finite output length n and other practical limitations of the considered shaping algorithm and reflects the resulting *rate loss*. The second term in (1) is the uncertainty resulting from the considered decoding metric q and the impairments introduced by the transmission over a channel with input X and output Y . An instance of (1) was first proposed in [41] and developed further in [42]. The uncertainty $\mathbb{U}(q, X, Y)$ is a generalization of the conditional entropy $\mathbb{H}(X|Y)$ of a memoryless channel with input X and output Y , in a similar spirit as the generalized mutual information (GMI) [43], [44] generalizes the mutual information $\mathbb{I}(X; Y)$. For uniform input, (1) recovers the GMI, while for shaped input, (1) is different from the GMI, see [42, Section 6.1]. We show how to calculate the uncertainty $\mathbb{U}(q, x^n, y^n)$ from channel measurements with input sequence x^n and output sequence y^n , and we provide the operational meaning of the uncertainty for general channels that lack an analytic description. In [45], the normalized generalized mutual information (NGMI) was proposed as a modification of the GMI tailored for PAS. Our expression (1) recovers the NGMI. In [46], various performance metrics are evaluated for shaped inputs. Instances of (1) are used in [47] to compare probabilistic and geometric shaping.

Outline: In Sec. II, we provide a detailed walk-through for the transmitter components of layered PS displayed in Fig. 1, explaining the basic principles of FEC encoding, PS encoding, and the integration of PS and FEC. In Sec. III, we explain a parameterization of the PAS transmitter as an instance of the layered PS architecture, we derive the size of the PAS shaping set and we relate it to achievable DM rates. In Sec. IV, we take an information-theoretic perspective and characterize achievable PS encoding rates for given and fixed FEC encoders. Again from an information-theoretic perspective, we then characterize in Sec. V an achievable FEC rate for which a transmitted sequence can be recovered from its noisy observation at the receiver. We derive the achievable FEC rate for a general decoding metric. In Sec. VI, we show that the results of Secs. IV and V recover the known quantities conditional entropy and mutual information when transmitting over memoryless channels. In Sec. VII we apply the developed framework to assess the performance in optical transmission experiments. In Sec. VIII, we conclude this work and give an outlook to future research directions.

Notation: We write row vectors as $(x_1, x_2, \dots, x_n) = x_1 \dots x_n = x^n = \mathbf{x}$ and use bold face letters, e.g., $\mathbf{x}$, when the number of entries is clear from the context. We denote matrices by capital bold letters, e.g., $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, and we denote the transpose of vectors and matrices by $(\cdot)^T$, e.g., $\mathbf{A}^T$. The symbols $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$ denote the set of real numbers and non-negative real numbers, respectively. We denote a random variable (RV) with capital letters, e.g., X , and their realizations with small letters, e.g., x . A discrete RV X has a distribution P_X on a discrete set $\mathcal{X}$, e.g., for $x \in \mathcal{X}$, $P_X(x) = \Pr(X = x)$. A continuous RV X takes real values and has density p_X , e.g., $\int_{-\infty}^x p_X(\tau) d\tau = \Pr(X \leq x)$. For a real-valued function

$f: \mathcal{X} \rightarrow \mathbb{R}$ and a discrete RV with distribution P_X on $\mathcal{X}$, the expectation of $f(X)$ is

$$\mathbb{E}[f(X)] = \sum_{x \in \text{supp}(P_X)} f(x) P_X(x) \quad (2)$$

where $\text{supp}(P_X) = \{x \in \mathcal{X} : P_X(x) > 0\}$ is the support of P_X . Note that the RVs appearing as functional arguments within the expectation operator identify the probability distribution over which the expectation is calculated. The entropy of a discrete distribution P_X is

$$\mathbb{H}(P_X) = \mathbb{H}(X) = \mathbb{E}[-\log_2 P_X(X)]. \quad (3)$$

The conditional entropy is

$$\mathbb{H}(X|Y) = \mathbb{E}[-\log_2 P_{X|Y}(X|Y)] \quad (4)$$

and the mutual information is

$$\mathbb{I}(X; Y) = \mathbb{H}(X) - \mathbb{H}(X|Y). \quad (5)$$

II. INTRODUCTION TO PROBABILISTIC SHAPING AND FEC CODING

In Fig. 1, we display a diagram of the layered PS architecture. In this section, we discuss the transmitter, i.e., the mapping of binary data u^k to shaped signals x^n at the channel input. We walk backwards through the layers: we first discuss higher-order modulation formats, then the mapping of bits to signal points, the FEC encoding of bits, and PS encoding, i.e., the mapping of bits to shaped sequences. We conclude by discussing the integration of PS and FEC encoding. Actual implementations often connect one or more of the transceiver components in Fig. 1 via interleavers. We discuss a specific length n_{fec} block interleaver for PS encoding in Example II.6. We comment on general length n block interleavers in Remark 5 and explain why their thorough study is beyond the scope of this paper. Interleaving beyond the block length n (e.g., by convolutional interleavers) adds additional latency, and such interleavers need to be carefully designed, which is beyond the scope of this work. We therefore omit such interleavers in our system model.

A. Higher-Order Modulation Formats

In polarization-division multiplexing (PDM) the coherent transmitter modulates data onto two polarizations, and in each polarization, the in-phase and quadrature components of the carrier are modulated separately. In each real dimension, we assume an M -amplitude shift keying (ASK) constellation with the constellation points $\{\pm 1, \pm 3, \dots, \pm(M-1)\}$, where M is even. A two-dimensional M^2 -quadrature amplitude modulation (QAM) constellation corresponds to the Cartesian product of two real-valued ASK constellations and a four-dimensional dual polarized QAM (DP-QAM) constellation corresponds to the Cartesian product of two QAM constellations and equivalently, to the Cartesian product of four ASK constellations. In the following, we denote the ASK constellation by $\mathcal{X}$ and the set of amplitudes by $\mathcal{A} = \{|x| : x \in \mathcal{X}\}$.

TABLE I
CONSTRUCTION OF PS ENCODER IN EXAMPLE II.7

$\{c = u_{\text{fec}}G\}$	$c \in S_b?$	u_{fec}	u	
0000000	✓	0000	000	PS encoder $u \mapsto u_{\text{fec}}$ 000 $\mapsto$ 0000 001 $\mapsto$ 0010 010 $\mapsto$ 0011 011 $\mapsto$ 0100 100 $\mapsto$ 0101 101 $\mapsto$ 1000 110 $\mapsto$ 1001 111 $\mapsto$ 1110
0001111	×			
0010110	✓	0010	001	
0011001	✓	0011	010	
0100101	✓	0100	011	
0101010	✓	0101	100	
0110011	×			
0111100	×			
1000011	✓	1000	101	
1001100	✓	1001	110	
1010101	×			
1011010	×			
1100110	×			
1101001	×			
1110000	✓	1110	111	
1111111	×			

B. Bit-Mapper

To use a binary FEC code with a higher-order modulation format (e.g., 2^m -ASK, or 2^m -QAM), each constellation point $x \in \mathcal{X}$ is labeled with m bits, i.e., $\chi : \mathcal{X} \rightarrow \{0, 1\}^m$ and $\chi(x) = b_1 \dots b_m = \mathbf{b}$. The inverse labeling function is $\chi^{-1} : \{0, 1\}^m \rightarrow \mathcal{X}$. The i th bit level of x is $b_i = \chi_i(x)$. The bit-mapper in Fig. 1 maps m consecutive bits to one constellation point, according to the labeling function χ . For ASK constellations, we also introduce the amplitude label $\chi_A : \mathcal{A} \rightarrow \{0, 1\}^{m-1}$.

C. Binary Linear Block Codes

We start by considering the transmitter without shaping, i.e., in Fig. 1, we have $u^k = u_{\text{fec}}^{k_{\text{fec}}}$.

1) *Parity Check Matrix*: To protect a block $\mathbf{c} = c_1 \dots c_{n_{\text{fec}}}$ of n_{fec} bits against errors, a linear FEC code imposes m_{fec} linear constraints on $\mathbf{c}$. Each constraint requires that a certain subset of the n_{fec} bits in $\mathbf{c}$ add to an even number, i.e., zeros in the binary field. The constraints are therefore called *parity checks*. The i th parity check is compactly written as a length n_{fec} row vector $\mathbf{h}_i = h_{i1} \dots h_{in_{\text{fec}}}$ and the vector $\mathbf{c}$ must fulfill $\mathbf{c}\mathbf{h}_i^T = 0$. Arranging m_{fec} parity checks in a matrix results in the *parity check matrix* $\mathbf{H}$ with transpose

$$\mathbf{H}^T = [\mathbf{h}_1^T \quad \mathbf{h}_2^T \quad \dots \quad \mathbf{h}_{m_{\text{fec}}}^T] \quad (6)$$

and $\mathbf{c}$ is a codeword if and only if it fulfills all m_{fec} parity checks, i.e., $\mathbf{c}\mathbf{H}^T = \mathbf{0}$. This defines the linear FEC code

$$\mathcal{C} := \{\mathbf{c} \in \{0, 1\}^{n_{\text{fec}}} : \mathbf{c}\mathbf{H}^T = \mathbf{0}\}. \quad (7)$$

Example II.1 ((7, 4) Hamming code): Consider the parity check matrix

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}. \quad (8)$$

The code length is $n_{\text{fec}} = 7$ and the parity check matrix $\mathbf{H}$ defines $m_{\text{fec}} = 3$ parity checks. The 16 codewords are listed in the first column of Table I.

2) *FEC Code Rate*: If the m_{fec} parity check equations are linearly independent, i.e., none of the parity checks $\mathbf{h}_i$ can be written as a sum of other parity checks $\mathbf{h}_{j \neq i}$, then $k_{\text{fec}} = n_{\text{fec}} - m_{\text{fec}}$ of the codeword entries can be filled with arbitrary information bits and the remaining m_{fec} entries are given by the parity checks. The number k_{fec} is the *information length* in bits, i.e., the *dimension* of the FEC code. The rate of a binary linear FEC code with alphabet $\mathcal{X} = \{0, 1\}$ and dimension k_{fec} is

$$R_{\text{fec}} = \frac{k_{\text{fec}}}{n_{\text{fec}}}. \quad (9)$$

The corresponding FEC overhead (OH) is

$$\text{OH}_{\text{fec}} = 100 \cdot \left(\frac{1}{R_{\text{fec}}} - 1 \right) \%. \quad (10)$$

Example II.2: The Hamming parity check matrix (8) defines $m_{\text{fec}} = 3$ independent parity checks on $n_{\text{fec}} = 7$ bits. Thus, the code dimension, rate, and overhead are

$$k_{\text{fec}} = 4, \quad R_{\text{fec}} = \frac{4}{7}, \quad \text{OH}_{\text{fec}} = 75\% \quad (11)$$

respectively.

We generalize the definition of the FEC code rate to general (not necessarily binary) alphabets $\mathcal{D}$ and general (not necessarily linear) codes as follows. A code $\mathcal{C} \subseteq \mathcal{D}^n$ of length n codewords with entries in $\mathcal{D}$ has FEC code rate

$$R_{\text{fec}} = \frac{\log_2 |\mathcal{C}|}{\log_2 |\mathcal{D}|^n} = \frac{\log_2 |\mathcal{C}|}{mn}, \quad m = \log_2 |\mathcal{D}|. \quad (12)$$

Note that the FEC code rate is bounded by $0 \leq R_{\text{fec}} \leq 1$ for any alphabet size $|\mathcal{D}|$. Note further that if $\mathcal{D}$ is binary and the code is linear, then definition (12) recovers (9).

3) *Systematic Encoding*: It is convenient for the last m_{fec} columns of $\mathbf{H}$ to be linearly independent, which can always be achieved, when $\mathbf{H}$ is full rank, by suitable rearrangement of columns. Then, the matrix is of the form

$$\mathbf{H} = [\mathbf{Q}|\mathbf{R}] \quad (13)$$

where the $m_{\text{fec}} \times m_{\text{fec}}$ matrix $\mathbf{R}$ is full rank and invertible. Systematic encoding of k_{fec} bits $\mathbf{u}$ can now be done in two steps.

- 1) Calculate $\mathbf{s} = \mathbf{u}\mathbf{Q}^T$.
- 2) Solve $\mathbf{p}\mathbf{R}^T = \mathbf{s} \Rightarrow \mathbf{p} = \mathbf{s}(\mathbf{R}^T)^{-1}$.

The vector $\mathbf{c} = [\mathbf{u}|\mathbf{p}]$ is then a codeword, i.e., it fulfills $\mathbf{c}\mathbf{H}^T = \mathbf{0}$. A convenient way to represent systematic encoding is via a systematic generator matrix

$$\mathbf{G} = [\mathbf{I}|\mathbf{P}] \quad (14)$$

where $\mathbf{I}$ is a $k_{\text{fec}} \times k_{\text{fec}}$ identity matrix and $\mathbf{P} = \mathbf{Q}^T(\mathbf{R}^T)^{-1}$. We can now compactly write systematic encoding by the multiplication of $\mathbf{u}$ with $\mathbf{G}$, i.e.,

$$\mathbf{u}\mathbf{G} = [\mathbf{u}|\mathbf{p}] = \mathbf{c}. \quad (15)$$

Since $\mathbf{p} = \mathbf{u}\mathbf{P}$, we call $\mathbf{P}$ the *parity forming part* of $\mathbf{G}$.

Example II.3: For the Hamming code parity check matrix, we have

$$\mathbf{H} = \left[\underbrace{\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{R}} \right] = [\mathbf{Q}|\mathbf{R}]. \quad (16)$$

Since $\mathbf{R}$ is an identity matrix, the parity forming part of the systematic generator matrix is $\mathbf{P} = \mathbf{Q}^T$ and $\mathbf{u}\mathbf{Q}^T = \mathbf{p}$ calculates the parity bits. For instance, for $\mathbf{u} = (1, 1, 0, 1)$, we get $\mathbf{p} = (0, 0, 1)$.

D. PS Encoding into Shaping Sets

We consider the transmitter without FEC encoding, i.e., in Fig. 1, $u_{\text{fec}}^{k_{\text{fec}}} = c^{n_{\text{fec}}}$. For a constellation $\mathcal{X}$ of size 2^m , the labeling function χ represents each entry of x^n by m bits. Consequently, $n_{\text{fec}} = mn$. The output of the PS encoder is the length mn binary sequence $c^{n_{\text{fec}}} = c^{mn}$, which takes values in a binary shaping set $\mathcal{S}_b \subseteq \{0, 1\}^{mn}$. The output of the bit-mapper is $\mathbf{x}$ in the shaping set $\mathcal{S} \subseteq \mathcal{X}^n$. The input $\mathbf{u}$ of the PS encoder indexes sequences in the *shaping set* and has length

$$k = \lfloor \log_2 |\mathcal{S}_b| \rfloor = \lfloor \log_2 |\mathcal{S}| \rfloor. \quad (17)$$

Note that in case the size of the shaping set is not a power of two, only a subset of the shaping set is actually used for transmission. We define the shaping set rate R_{ps} as the ratio of the number of bits we need for addressing an element in the shaping set and the number of bits we need to address the corresponding symbol in the n -dimensional constellation $\mathcal{X}^n$, i.e.,

$$R_{\text{ps}} = \frac{\lfloor \log_2 |\mathcal{S}| \rfloor}{\log_2 |\mathcal{X}|^n} = \frac{\lfloor \log_2 |\mathcal{S}| \rfloor}{mn} = \frac{k}{mn}. \quad (18)$$

Note that $0 \leq R_{\text{ps}} \leq 1$. We introduce the PS encoder by discussing on-off-keying (OOK) shaping sets for non-coherent transmission and 4-ASK shaping sets for coherent transmission.

1) *Non-Coherent Transceiver:* In non-coherent transmission, intensity modulation (IM) is used, i.e., the intensity (power) of the optical signal is modulated, which is proportional to the amplitude of the electrical signal [48], [49]. The simplest form of IM is OOK. When a bit is 0, the power is switched off and when the bit is 1, the power is Δ , i.e., the bit-mapper implements the mapping $0 \mapsto 0, 1 \mapsto \Delta$. Consider now a length n sequence with n_0 zeros and n_1 ones. The average power is

$$\frac{n_0}{n} \cdot 0 + \frac{n_1}{n} \cdot \Delta = \frac{n_1}{n} \cdot \Delta. \quad (19)$$

Following [50], suppose we encounter the following three system design requirements:

- 1) The on-power Δ should be large, so that the detector at the receiver can reliably distinguish between Δ (“on”) and 0 (“off”).
- 2) The average power should be low.
- 3) The rate in bits/OOK symbol should be large.

The first two requirements are in conflict: increasing Δ also increases the average power. We resolve this conflict by backing-off in rate, i.e., we use for transmission only power-efficient sequences with a limited number of ones.

Example II.4 (Shaping set for OOK): We transmit length $n = 7$ sequences with “on” at most 3 times, i.e., we consider as binary shaping set $\mathcal{S}_b$ the sequences with either no 1, one 1, two 1s, or three 1s. The resulting shaping set $\mathcal{S}$ has cardinality

$$\begin{aligned} |\mathcal{S}| = |\mathcal{S}_b| &= \binom{7}{0} + \binom{7}{1} + \binom{7}{2} + \binom{7}{3} \\ &= 1 + 7 + 21 + 35 = 64. \end{aligned}$$

We index these sequences by

$$k = \lfloor \log_2 |\mathcal{S}_b| \rfloor = 6 \text{ bits} \quad (20)$$

which corresponds to the rate

$$\frac{k}{n} = \frac{\lfloor \log_2 |\mathcal{S}| \rfloor}{n} = \frac{6}{7} \left[\frac{\text{bits}}{\text{OOK symbol}} \right]. \quad (21)$$

Remark 1: Shell mapping [35], [51] and enumerative coding [29], [52] can efficiently index the lowest energy sequences of OOK.

Let us compare the shaped OOK in Example II.4 to a conventional system that operates at the same average power P . The conventional system transmits any length 7 sequence, has rate $7/7 = 1$ bit/OOK symbol, and instantaneous power $\Delta = 2P$. Our shaped OOK has rate $6/7$ bit/OOK symbol, and instantaneous power

$$\Delta = \frac{64 \cdot 7}{7 \cdot 1 + 21 \cdot 2 + 35 \cdot 3} P = 2.9P. \quad (22)$$

Thus, shaped OOK has a lower rate, but its instantaneous power Δ is larger, so that the receiver can distinguish 0 and Δ more reliably. This observation is fundamental: shaping is coding that trades rate for reliability via the instantaneous power.

2) *Coherent Transceiver:* In coherent transmission, the power of the optical signal is proportional to the square of the electrical signal and we modulate both sign and amplitude. Consider a system using the 4-ASK constellation

$$\mathcal{X} = \{\pm 1, \pm 3\}. \quad (23)$$

Let x^n be a length n sequence with entries in $\mathcal{X}$ that is scaled by Δ and transmitted. The average power is

$$\frac{\sum_{i=1}^n |x_i|^2 \Delta^2}{n}. \quad (24)$$

Two key observations are

- 1) The signs of the signal points have no influence on the average power (24).
- 2) The average power (24) decreases when signal points with small amplitude are used more often than signal points with large amplitude.

Example II.5 (Shaping set for 4-ASK): Consider the transmission of 4-ASK sequences of length $n = 4$. We represent such a sequence by a sequence $\mathbf{a}$ of amplitudes in $\{1, 3\}$ and a sequence $\mathbf{s}$ of signs in $\{-1, +1\}$. Since the sign does not change the instantaneous power, we use all possible sign sequences. We use low power sequences where the amplitude 3 occurs at most

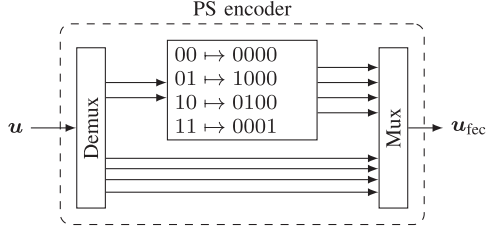


Fig. 2. Realization of a PS encoder. Two bits are encoded to 4 bits of low average power. 4 additional bits are bypassed. In total, the PS encoder has 6 input bits and 8 output bits.

once. The shaping set is thus

$$\begin{aligned} \mathcal{S} = \{ & (\pm 1, \pm 1, \pm 1, \pm 1), (\pm 3, \pm 1, \pm 1, \pm 1), \\ & (\pm 1, \pm 3, \pm 1, \pm 1), (\pm 1, \pm 1, \pm 3, \pm 1), \\ & (\pm 1, \pm 1, \pm 1, \pm 3) \} \end{aligned} \quad (25)$$

which is of size

$$|\mathcal{S}| = \left[\binom{4}{0} + \binom{4}{1} \right] \cdot 2^4 = 80. \quad (26)$$

Since 80 is not a power of two, we index by $\lfloor \log_2 |\mathcal{S}| \rfloor = 6$ bits the sequences in $\mathcal{S}$ of lowest power. The resulting rate is

$$\frac{\lfloor \log_2 |\mathcal{S}| \rfloor}{n} = \frac{6}{4} \left[\frac{\text{bits}}{\text{ASK symbol}} \right]. \quad (27)$$

Remark 2: Extending Remark 1, we can use shell-mapping or enumerative coding for indexing the lowest power amplitude sequences and we can index the sign sequences by n additional bits, e.g., by the mapping $0 \mapsto -1, 1 \mapsto +1$.

Example II.6: We construct a shaping encoder for 4-ASK with the labeling function χ given by

$$\begin{array}{ccccccc} x & -3 & -1 & +1 & +3 \\ \hline \chi(x) & 11 & 01 & 00 & 10 \end{array}$$

We use the labeling function χ and an interleaver f to represent a length n sequence x^n by a binary length $2n$ sequence via

$$x_1 \dots x_n \xrightarrow{\chi, f} (\chi_1(x_1), \dots, \chi_1(x_n), \chi_2(x_1), \dots, \chi_2(x_n)) \quad (28)$$

that is, we first list all bit-level 1 values and then all bit-level 2 values. We can now identify the shaping set $\mathcal{S}$ with its binary representation via $\mathcal{S}_b = f(\chi(\mathcal{S}))$. By the mapping (28), we see that the first 4 bits of the binary representation identify the amplitude and the last 4 bits identify the sign. Note that the 4 sign bits are unconstrained and the 4 amplitude bits cannot be chosen freely, so we split the shaping encoder in two parts. For the 4 constrained bits there are 5 permitted amplitude sequences (see (25)), of which an encoder picks the 4 sequences of lowest power and indexes them by two bits. The four unconstrained bits can be bypassed without further processing. We display the complete PS encoder in Fig. 2.

E. Layered Probabilistic Shaping for OOK

We integrate PS encoding with linear FEC encoding and we first consider OOK. The requirement for c at the FEC output is

- 1) c is in the shaping set $\mathcal{S}_b$.
- 2) c is in the linear FEC code $\mathcal{C}$.

Thus, to encode a message, we have to map it to a vector in the intersection $\mathcal{C} \cap \mathcal{S}_b$ of the shaping set $\mathcal{S}_b$ and the linear FEC code $\mathcal{C}$. We accomplish this with a PS encoder between the data source and the FEC encoder. The PS encoder ensures that the codewords that are output by the FEC encoder map to signals in the shaping set. We construct the shaping code as follows:

- 1) List all codewords in the FEC code.
- 2) Identify the codewords that are in $\mathcal{S}_b \cap \mathcal{C}$.
- 3) Identify the systematic part of these codewords as it indexes the codewords.
- 4) Choose $2^{\lfloor \log_2 |\mathcal{S}_b \cap \mathcal{C}| \rfloor}$ systematic parts and create a lookup table (LUT) indexed by $\lfloor \log_2 |\mathcal{S}_b \cap \mathcal{C}| \rfloor$ bits.

Example II.7: We combine the OOK shaping set of Example II.4 with the binary linear FEC code of Example II.1. In Table I, we show the construction of the shaping code and display the resulting LUT that implements the PS encoder. The PS encoder has rate $R_{ps} = \frac{3}{4}$. The PS-FEC rate is

$$R_{ps} R_{fec} = \frac{3}{4} \cdot \frac{4}{7} = \frac{3}{7}. \quad (29)$$

The key problem of this naive implementation is the shaping LUT, which requires memory exponentially large in k . Future research hopefully finds an efficient algorithm replacing the shaping LUT. Under certain conditions, namely when only a part of the coded bits are required to be shaped, a method for efficient indexing is known. We discuss this next.

F. Probabilistic Amplitude Shaping for Coherent Transmission

We now integrate shaped ASK with linear FEC coding.

Example II.8 (Probabilistic Amplitude Shaping): Consider the system setup of Example II.6 including the bit-mapper and the interleaver. The 4-ASK shaping set (25) of Example II.5 has in its binary representation $\mathcal{S}_b$ length-8 bit sequences. We want to transmit sequences that are both in the shaping set $\mathcal{S}_b$ and in a binary code $\mathcal{C}$. We use a single parity check (SPC) code with the parity check matrix

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad (30)$$

so we have $n_{fec} = 8$, $m_{fec} = 1$, and $k_{fec} = 7$. We could construct the PS encoder by identifying the messages that map to vectors in $\mathcal{C} \cap \mathcal{S}_b$ in the same way as we did in Example II.7. We choose a different approach. From the FEC code perspective, we can choose 7 bits freely and 1 bit is constrained by the parity check (30). From the shaping set perspective, we can choose the 4 sign bits freely and the 4 amplitude bits are shaped. To integrate PS and FEC, we modify the PS encoder of Example II.6 by reserving one of the sign bits for the parity bit. Thus, as displayed in Fig. 3, 2 information bits are mapped to 4 amplitude bits, 3 information bits are mapped to 3 sign bits, and the FEC code calculates 1 parity bit, which is mapped to the remaining sign

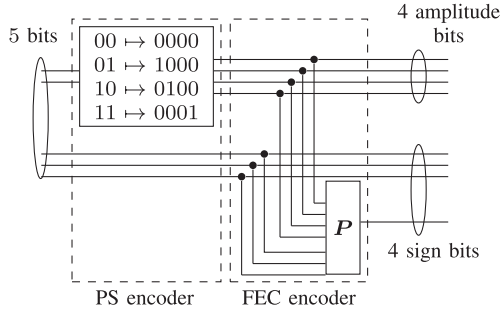


Fig. 3. The PAS encoder of Example II.6.

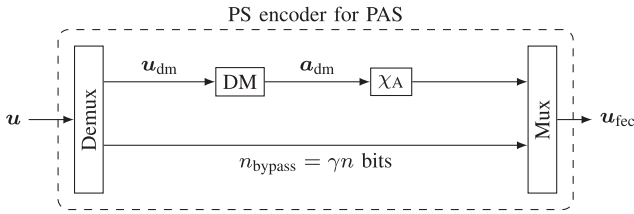


Fig. 4. The PS encoder for PAS (1-dimensional constellation).

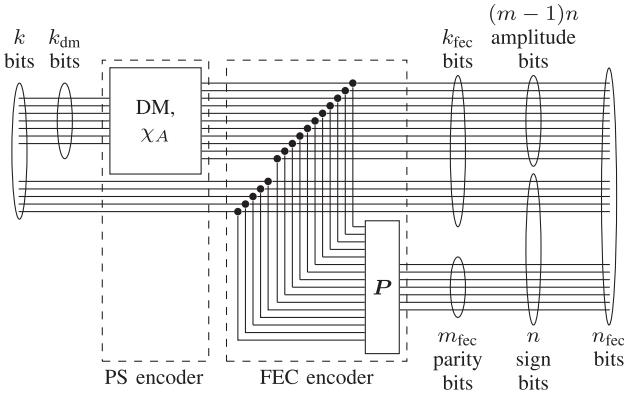


Fig. 5. The PAS encoder (1-dimensional constellation).

bit. This is an example of PAS, for which we provide a detailed description in III-A. We distinguish three rates:

- 1) The FEC code rate is $R_{\text{fec}} = 7/8$.
- 2) The PS rate is $R_{\text{ps}} = 5/7$.
- 3) The PS-FEC rate is $R_{\text{ps}}R_{\text{fec}} = 5/8$.

III. PAS TRANSMITTER

In this section, we detail the PAS architecture that we introduced in Example II.8.

A. PAS Encoding

Recall that a PS encoder converts a length k bit sequence u into a bit sequence u_{fec} of length k_{fec} , which is then FEC encoded into a codeword c in the shaping set, i.e., $c \in \mathcal{S}_b \cap \mathcal{C}$. While in general, the naive PS encoder requires a prohibitively large LUT (see Example II.7), a PAS encoder achieves the PS-FEC encoding efficiently. The PAS encoder is displayed in Fig. 5 and the PS encoder component is detailed in Fig. 4. The key component of the PS encoder is a DM, which maps k_{dm} bits to

a length n shaped amplitude sequence a_{dm} . We therefore define the DM rate

$$R_{\text{dm}} = \frac{k_{\text{dm}}}{n} \quad (31)$$

in bits per amplitude. For 1-dimensional constellations, those n amplitudes are then mapped to $(m-1)n$ bits by the amplitude label function χ_A . The amplitude of an N -dimensional signal point from an N -dimensional constellation with 2^m signal points is mapped to $m-N$ bits accordingly. Consequently, the DM rate is bounded by

$$0 \leq R_{\text{dm}} \leq m - N. \quad (32)$$

In Fig. 5, we can see that for 1-dimensional constellations, PAS requires that the number of parity bits m_{fec} does not exceed the number n of sign bits. In other words, the systematic encoding of the bits representing the amplitudes requires to have $k_{\text{fec}} \geq (m-1)n$. Using the relation (9) and $n_{\text{fec}} = mn$, we get the condition

$$R_{\text{fec}} = \frac{k_{\text{fec}}}{n_{\text{fec}}} = \frac{k_{\text{fec}}}{mn} \geq \frac{(m-1)n}{mn} = \frac{m-1}{m}. \quad (33)$$

A signal point of an N -dimensional constellation has N signs, so for an N -dimensional constellation with 2^m signal points, the PAS condition becomes

$$R_{\text{fec}} \geq \frac{m-N}{m}. \quad (34)$$

Remark 3: If the PAS condition (33) is not fulfilled, power efficient signaling with layered PS can still be achieved by using the more general encoding that we exemplified for OOK in II-E. For example, in [16], rates were reported that are achievable by layered PS, but not by PAS, since the condition (33) is not fulfilled. In [50], PS for pulse amplitude modulation is considered, where the signs cannot be modulated and PAS cannot be used for the considered exponential distribution on the amplitudes. In Sec. II-D, we show achievability of PS encoding for the general case by information-theoretic arguments. However, besides encoding by large LUTs as exemplified in Example II.7, we currently do not know a method to efficiently implement PS encoding in the general case. The benefits of PS reported in [16], [50] motivate future research on finding such a method.

B. PAS Rate Taxonomy

For the parametrization of the PAS rate, it is convenient to define the fraction γ of sign bits that are used for data bits. It is given by

$$\gamma = \frac{n - m_{\text{fec}}}{n}. \quad (35)$$

Note that γ does not depend on the constellation dimension N . In Example II.8, we have 4 sign bits and 1 parity bit, which corresponds to $\gamma = 3/4$. We instantiate the relevant rates R_{ps} , R_{fec} and the resulting spectral efficiency (SE) η for different signaling schemes in Table II. We distinguish one, two and N -dimensional schemes. For 2^m -ASK, we use the rate parametrization in the first row of Table II. When the I and Q components of a 2^m -QAM constellation are processed jointly, using the QAM parametrization in row two of Table II is more convenient. For

TABLE II
PAS RATES

	R_{ps}	R_{fec}	$\eta = mR_{ps}R_{fec}$
2^m -ASK	$\frac{R_{dm} \cdot (m-1) + \gamma}{m-1+\gamma}$	$\frac{m-1+\gamma}{m}$	$R_{dm}(m-1) + \gamma$
2^m -QAM	$\frac{R_{dm} \cdot (m-2) + 2\gamma}{m-2+2\gamma}$	$\frac{m-2+2\gamma}{m}$	$R_{dm}(m-2) + 2\gamma$
2^m -ND	$\frac{R_{dm} \cdot (m-N) + N\gamma}{m-N+N\gamma}$	$\frac{m-N+N\gamma}{m}$	$R_{dm}(m-N) + N\gamma$

high-dimensional modulation formats, e.g., [19], [26], the N -dimensional parametrization in row three of Table II may be a good choice.

C. Shaping Set Size

We now discuss the shaping set size for 1-dimensional constellations. The shaping set $\mathcal{S}$ for PAS results from combining each amplitude sequence in the set $\mathcal{S}_A$ of shaped amplitude sequences with all 2^n sign sequences. The resulting shaping set size is

$$|\mathcal{S}| = 2^n |\mathcal{S}_A|. \quad (36)$$

If a specific DM is used, it is convenient to define the amplitude shaping set $\mathcal{S}_A$ as the set formed by the output sequences of the DM, in which case it contains exactly $|\mathcal{S}_A| = 2^{k_{dm}}$ sequences and the size of the shaping set becomes

$$|\mathcal{S}| = 2^n \cdot 2^{k_{dm}}. \quad (37)$$

The number of bits per symbol required to index the sequences in $\mathcal{S}$ is then

$$\frac{\log_2 |\mathcal{S}|}{n} = 1 + \frac{\log_2 |\mathcal{S}_A|}{n} = 1 + \frac{k_{dm}}{n}. \quad (38)$$

D. Achievable DM Rates

The letter distribution of the amplitude shaping set is

$$\hat{P}_A(a) = \frac{1}{|\mathcal{S}_A|} \sum_{\mathbf{a}_{dm} \in \mathcal{S}_A} \frac{N(a|\mathbf{a}_{dm})}{n}, \quad a \in \mathcal{A} \quad (39)$$

where $N(a|\mathbf{a}_{dm})$ is the number of occurrences of amplitude a in sequence $\mathbf{a}_{dm}$. With $\hat{P}_A(a)$ we can conveniently calculate the average energy of the shaping set per symbol by

$$\mathbb{E}[X^2] = \mathbb{E}[A^2] = \sum_{a \in \mathcal{A}} \hat{P}_A(a) \cdot a^2. \quad (40)$$

The DM rate is upper bounded by the entropy $\mathbb{H}(\hat{P}_A)$ and we define the *rate loss* by

$$R_{loss} = \mathbb{H}(\hat{P}_A) - R_{dm}. \quad (41)$$

There are asymptotically optimal DMs for which the rate loss vanishes for large block lengths [3]. For dyadic distributions, there are short variable length DMs with zero rate loss [53]–[55].

E. Constant Composition Distribution Matching (CCDM)

To exemplify how DM rates can be calculated, we instantiate the DM by constant composition distribution matching (CCDM) as introduced in [3]. Every sequence output by a CCDM is a

permutation of some fixed sequence $\mathbf{a}$. Since every sequence is composed of exactly the symbols that occur in $\mathbf{a}$, we call this a constant composition codebook. The set of all permutations $\mathcal{T}_{\mathbf{a}}$ of the sequence $\mathbf{a}$ is

$$\mathcal{T}_{\mathbf{a}} = \{\mathbf{v} \in \mathcal{A}^n \mid N(a|\mathbf{v}) = N(a|\mathbf{a}), \quad a \in \mathcal{A}\}, \quad (42)$$

where $\mathcal{A}$ is the alphabet of amplitude values. The number of distinct permutations is given by the multinomial

$$|\mathcal{T}_{\mathbf{a}}| = \frac{n!}{\prod_{a \in \mathcal{A}} N(a|\mathbf{a})!}. \quad (43)$$

The CCDM indexes $2^{k_{dm}}$ sequences in $\mathcal{T}_{\mathbf{a}}$ where

$$k_{dm} = \lfloor \log_2 |\mathcal{T}_{\mathbf{a}}| \rfloor. \quad (44)$$

The rate loss of CCDM vanishes [3], i.e.,

$$\lim_{n \rightarrow \infty} \frac{k_{dm}}{n} = \mathbb{H}(\hat{P}_A). \quad (45)$$

IV. INFORMATION THEORY: FEASIBLE PS-FEC RATE

We now relate the FEC rate R_{fec} and the shaping set size $|\mathcal{S}|$ to the PS rate R_{ps} at which successful encoding into the intersection of shaping set $\mathcal{S}$ and FEC code $\mathcal{C}$ is possible.

To keep the notational overhead low, we interpret in this and the next section the FEC code as defined directly on the constellation $\mathcal{X}$ so that the codewords are sequences consisting of n signal points in $\mathcal{X}$. Consistent with (12), we analyze the size of the FEC code $\mathcal{C} \subset \mathcal{X}^n$ in terms of the FEC rate

$$R_{fec} = \frac{\log_2 |\mathcal{C}|}{nm}, \quad m = \log_2 |\mathcal{X}| \quad (46)$$

bounded by $0 \leq R_{fec} \leq 1$.

A. FEC Code Ensemble

To show that FEC codes allowing for successful PS encoding do exist, we use the information-theoretic tool called *random coding*: instead of looking at a specific FEC code, we analyze the probability of successful PS encoding for an FEC code picked at random. We consider the random code ensemble

$$\mathcal{C} = \{C^n(w), w = 1, 2, \dots, 2^{nmR_{fec}}\} \quad (47)$$

where the entries of the $|\mathcal{C}| = 2^{nmR_{fec}}$ codewords are chosen independently and uniformly distributed on the constellation $\mathcal{X}$.

We now generalize the shaping code construction exemplified in Table I to the random code ensemble. We consider a general shaping set $\mathcal{S} \subseteq \mathcal{X}^n$. For a PS rate R_{ps} with

$$0 \leq R_{ps} \leq 1 \quad (48)$$

we divide the FEC code into $2^{nmR_{ps}R_{fec}}$ partitions, so that the number of codewords in each partition is

$$\frac{2^{nmR_{fec}}}{2^{nmR_{ps}R_{fec}}} = 2^{nm(1-R_{ps})R_{fec}}. \quad (49)$$

The PS encoder maps message $u \in \{1, 2, \dots, 2^{nmR_{ps}R_{fec}}\}$ to a codeword in the u th partition that is in $\mathcal{S}$. An encoding error occurs if the PS encoder cannot find such a codeword.

Theorem 1: The probability that the PS encoder cannot map its input to a codeword in $\mathcal{C} \cap \mathcal{S}$ is upper bounded by

$$\Pr(\text{PS encoding failure}) \leq \exp \left(-2^{nm} \left[\frac{\log_2 |\mathcal{S}|}{mn} - (1 - R_{\text{fec}}) - R_{\text{ps}} R_{\text{fec}} \right] \right). \quad (50)$$

Proof: We prove this theorem in Appendix A. ■
For

$$R_{\text{ps}} R_{\text{fec}} < \frac{\log_2 |\mathcal{S}|}{mn} - (1 - R_{\text{fec}}) \quad (51)$$

the inner exponent in (50) is positive and the right-hand side of (50) approaches zero doubly exponentially fast in n . For a given FEC rate $R_{\text{fec}} > 0$, we divide the right-hand side of (51) by R_{fec} and we define the achievable PS rate by

$$R_{\text{ps}}^* = \left[1 - \frac{1}{R_{\text{fec}}} \left(1 - \frac{\log_2 |\mathcal{S}|}{mn} \right) \right]^+. \quad (52)$$

For $0 \leq R_{\text{ps}} < R_{\text{ps}}^*$, the inner exponent in (50) is positive and for large n , if R_{ps} is just a little bit smaller than R_{ps}^* , then the error bound is already very close to zero.

B. Asymptotic PS Rate: Constant Composition Shaping Set

For a distribution P_X , define $\mathcal{S}_n$ as the set of length n sequences $\mathbf{x}$ where letter v appears *approximately* $P_X(v) \cdot n$ times, for each letter $v \in \mathcal{X}$, in the sense that

$$\left| P_X(v) - \frac{N(v|\mathbf{x})}{n} \right| \xrightarrow{n \rightarrow \infty} 0. \quad (53)$$

Such approximations can be achieved, e.g., with the quantization algorithms proposed in [56]. Then

$$\lim_{n \rightarrow \infty} \frac{\log_2 |\mathcal{S}_n|}{n} = \mathbb{H}(P_X) \quad (54)$$

and the achievable PS rate for asymptotically large n is

$$R_{\text{ps}}^* = \left[1 - \frac{1}{R_{\text{fec}}} \left(1 - \frac{\mathbb{H}(P_X)}{m} \right) \right]^+. \quad (55)$$

Multiplying by R_{fec} , we have

$$R_{\text{ps}}^* R_{\text{fec}} = \left[\frac{\mathbb{H}(P_X)}{m} - (1 - R_{\text{fec}}) \right]^+. \quad (56)$$

V. INFORMATION THEORY: ACHIEVABLE FEC RATES

We now consider the sub-system of Fig. 1 starting from the FEC encoder output x^n to the FEC decoder input y^n , which is a noisy observation of x^n . We relate the FEC rate to the question of whether the decoder can successfully detect x^n from y^n . Intuitively, when the FEC code contains only two codewords, we are very likely to detect x^n from y^n . On the other hand, if the FEC code is very large, for example, if it contains all possible sequences without constraints, then we are very likely to detect a wrong sequence.

As in the previous section, to keep the notational overhead low, we interpret the FEC code as defined directly on the constellation $\mathcal{X}$ so that the codewords are sequences consisting of n signal points in $\mathcal{X}$.

A. FEC Decoder

We consider a generic FEC decoder with a decoding metric q . For an observation y^n , the metric assigns to each sequence $x^n \in \mathcal{X}^n$ a non-negative score $q(x^n, y^n)$. The FEC encoder maps a message w to a codeword $c^n(w)$. For an observed output y^n , the decoder outputs as its decision the message that maps to the codeword with the maximum score, i.e.,

$$\hat{w} = \underset{w \in \{1, \dots, |\mathcal{C}|\}}{\operatorname{argmax}} q(c^n(w), y^n). \quad (57)$$

Note that the decoder interprets x^n and y^n to be “close” when $q(x^n, y^n)$ is large, so the decoding metric q is not a metric in the mathematical sense. The terminology “decoding metric” is common in the coding literature, a famous example is the Fano metric, see, e.g., [57, Sec. 7.1].

Note that the decoder ignores the possible existence of a PS encoder in the sense that it *evaluates the metric over the entire codebook $\mathcal{C}$ and not over $\mathcal{C} \cap \mathcal{S}$* .

We explicitly allow the metric q to assign the score 0 to impossible input-output (I/O) pairs. For example, consider a noiseless binary symmetric channel (BSC) with input X , output Y , and crossover probability $\delta = 0$. We can describe it by the transition probabilities

$$P_{Y|X}(0|0) = P_{Y|X}(1|1) = 1 \quad (58)$$

$$P_{Y|X}(1|0) = P_{Y|X}(0|1) = 0. \quad (59)$$

A decoding metric for this channel is $q(x, y) = P_{Y|X}(y|x)$, which assigns score 0 to the impossible I/O pairs (0,1) and (1,0) and score 1 to the possible I/O pairs (0,0) and (1,1). In the proof of the following Theorem 2 (see Appendix A), we will need to divide by $q(x^n, y^n)$ for a *possible* I/O pair. For this technical reason, we define a non-negative decoding metric as a metric that assigns non-negative scores (including score 0) to impossible I/O pairs and strictly positive scores to possible I/O pairs. This definition is consistent with using channel transition probabilities as decoding metric.

B. Uncertainty and the Probability of Erroneous Decoding

Theorem 2: Suppose codeword $C^n(w_0) = x^n$ is transmitted, let y^n be a channel output sequence, and let q be a non-negative decoding metric. Define the *uncertainty*

$$\mathbb{U}(q, x^n, y^n) = -\frac{1}{n} \log_2 \frac{q(x^n, y^n)}{\sum_{v^n \in \mathcal{X}^n} q(v^n, y^n)}. \quad (60)$$

The probability that the decoder (57) does not recover the index w_0 from the sequence y^n is bounded from above by

$$\Pr(\hat{W} \neq w_0 | C^n(w_0) = x^n, Y^n = y^n) \leq 2^{-nm \left(1 - \frac{\mathbb{U}(q, x^n, y^n)}{m} - R_{\text{fec}} \right)}. \quad (61)$$

Proof: See Appendix B. ■

Based on (61), we define the achievable FEC rate

$$R_{\text{fec}}^*(q, x^n, y^n) = \left[1 - \frac{\mathbb{U}(q, x^n, y^n)}{m} \right]^+. \quad (62)$$

The bound (61) approaches zero exponentially fast in the rate backoff $R_{\text{fec}}^*(q, x^n, y^n) - R_{\text{fec}}$. For $R_{\text{fec}} \geq R_{\text{fec}}^*(q, x^n, y^n)$, the

bound is larger or equal to 1, which is a trivial upper bound on the error probability. Thus, the term $R_{\text{fec}}^*(q, x^n, y^n)$ defines a phase transition for R_{fec} towards reliable decoding. We can write (62) also in terms of the redundancy of the FEC code. We have

$$[1 - R_{\text{fec}}^*(q, x^n, y^n)] m = \mathbb{U}(q, x^n, y^n) \quad (63)$$

that is, the phase transition towards successful decoding occurs when the transmitted redundancy equals the uncertainty.

Remark 4 (Interpretation): Note that in Theorem 2, the transmitted codeword x^n and the noisy observation y^n are *deterministic* values. The code whose decoder attempts to recover x^n from y^n is *random*, i.e., the theorem bounds the probability of picking a bad code. This corresponds to what we observe when designing FEC codes in practice: with a large rate backoff $R_{\text{fec}}^*(q, x^n, y^n) - R_{\text{fec}}$, it is easy to find a practical rate R_{fec} code that decodes correctly, corresponding to a vanishing probability of picking a bad code. The larger R_{fec} is (i.e., the smaller the backoff becomes), the more difficult it is to find a practical rate R_{fec} code that decodes correctly and has desired properties such as, e.g., a small decoding threshold [58] or a minimum distance that grows linearly with the block length [59].

By making the achievable FEC rate $R_{\text{fec}}^*(q, x^n, y^n)$ dependent on the input sequence x^n and the output sequence y^n , Theorem 2 holds in great generality and attaches a precise operational meaning to the uncertainty for arbitrary channels. Consider now a binary channel and the decoding metric

$$q(x^n, y^n) = 2^{-d_H(x^n, y^n)} \quad (64)$$

where d_H is the Hamming distance, which counts the number of entries in which x^n and y^n differ. We calculate the uncertainty and the achievable FEC rate for various sequence pairs:

$$\mathbb{U}(q, 000, 000) = 0.585, \quad R_{\text{fec}}^*(q, 000, 000) = 0.415 \quad (65)$$

$$\mathbb{U}(q, 000, 100) = 0.918, \quad R_{\text{fec}}^*(q, 000, 100) = 0.082 \quad (66)$$

$$\mathbb{U}(q, 000, 110) = 1.25, \quad R_{\text{fec}}^*(q, 000, 110) = 0. \quad (67)$$

We observe that, depending on the received output sequence, the achievable FEC rate may be 0.415, 0.08, or 0. Thus, we need to choose an FEC rate that is achievable for a sufficient fraction of sequence pairs that occur during transmission and this FEC rate then provides an achievable SE. We derive an achievable SE for memoryless channels in Sec. VI-D by invoking the weak law of large numbers (WLLN), and we discuss the estimation of SEs for more general channels in VII-G.

C. Minimizing the Uncertainty

Theorem 2 provides via the uncertainty an achievable FEC code rate. We want this achievable FEC rate to be as large as possible and equivalently, we want the uncertainty to be small. First, we should choose an appropriate decoding metric q , which we discuss in more detail in Sec. VI. Second, we can tighten the bound in (61) as follows. Consider two metrics q and q' that lead for the decoding rule (57) to the same decoding decision, i.e., for any two codewords $a^n, b^n \in \mathcal{C}$,

$$q(a^n, y^n) > q(b^n, y^n) \Leftrightarrow q'(a^n, y^n) > q'(b^n, y^n). \quad (68)$$

In other words, exchanging q with q' preserves the order that decoding rule (57) imposes on the codewords. Then, the exact error probability on the left-hand side of (61) is equal for q and q' . However, the uncertainties $\mathbb{U}(q)$ and $\mathbb{U}(q')$ are in general not equal, since the uncertainty is the result of several simplifying bounding steps, see the derivation of the bound in Appendix B. For a fixed and given metric q , an order preserving transformation is

$$s > 0, \quad q'(x^n, y^n) = q(x^n, y^n)^s. \quad (69)$$

To tighten the bound (61), we minimize the uncertainty over s and we define

$$\mathbb{U}^*(q, x^n, y^n) := \min_{s > 0} \mathbb{U}(q^s, x^n, y^n). \quad (70)$$

A similar optimization over a parameter s can be found in the definition of the GMI in [43].

Example V.1: As suggested in [60, Example 2], we can interpret the optimization over s in (70) as a model parameter estimation. Consider the transmission over an AWGN channel where the noise variance is not known exactly so that the decoder resorts to an estimate $\tilde{\sigma}^2$ and uses the decoding metric

$$\tilde{q}(x, y; \tilde{\sigma}^2) = \exp\left(-\frac{(y - x)^2}{2\tilde{\sigma}^2}\right). \quad (71)$$

Considering now $\tilde{q}(x, y)^s$ and minimizing the corresponding uncertainty over s yields an optimized s^* that provides the refined variance estimate $\sigma^2 = \tilde{\sigma}^2 / s^*$. In other words, the optimization over s compensates for a model mismatch.

D. Decoding Metrics in the Log-Domain

In particular for binary FEC codes, it is convenient to use decoding metrics that can take positive and negative values, for example, log-likelihood ratios (LLRs). The transformation $q_{\log} \mapsto e^{q_{\log}}$ leads to a non-negative metric implementing the same decoding rule, so that we can calculate the uncertainty for a log-domain metric $q_{\log}$ by evaluating (60) in $e^{q_{\log}}$ and we calculate the minimized uncertainty by

$$\mathbb{U}^*(e^{q_{\log}}, x^n, y^n) := \min_{s > 0} \mathbb{U}(e^{sq_{\log}}, x^n, y^n). \quad (72)$$

We show an instance of the metric $q_{\log}$ in Sec. VI-C.

E. Memoryless Metrics

Practical decoding metrics are often of the form

$$q^n(x^n, y^n) = \prod_{i=1}^n q(x_i, y_i) \quad (73)$$

which is called *memoryless*, since the individual score $q(x_i, y_i)$ assigned to codeword entry x_i and its noisy observation y_i does not depend on other codeword entries and observations. For memoryless metrics, the uncertainty (62) becomes

$$\mathbb{U}(q, x^n, y^n) = -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{q(x_i, y_i)}{\sum_{v \in \mathcal{X}} q(v, y_i)}. \quad (74)$$

Remark 5 (Interleaving does not change the achievable FEC rate): From (73), we can see that the value of the metric does not

change when we use a length n block interleaver between the channel and the FEC components. Consequently, the uncertainty does not change either, so the achievable FEC code rate cannot be improved by interleaving. The reason for this is twofold. First, the uncertainty is calculated assuming the maximum-likelihood (ML)-like decoder (57). Second, the achievable FEC code rate is stated in terms of the probability of picking a code uniformly at random that achieves it. The selection of a code uniformly at random has the effect of a random interleaver between the FEC components and the channel, and consequently, any effect of choosing a specific interleaver is undone by the random coding experiment. When we design practical FEC codes with practical decoding, e.g., low-density parity-check (LDPC) codes with iterative decoding, then the interleaver can indeed improve the performance, see, e.g., [36], [61]. FEC codes and block interleavers should therefore be designed jointly. This, however, is beyond the scope of this paper.

VI. UNCERTAINTY AND ACHIEVABLE SPECTRAL EFFICIENCIES FOR MEMORYLESS CHANNELS

In this section, we consider memoryless channels and we extend the results to more general channels in Sec. VII-G.

A. Asymptotic Uncertainty for Memoryless Channel

For memoryless channels $p_{Y|X}$ (e.g., the additive white Gaussian noise (AWGN) channel) and large n , we can identify uncertainty (60) with a conditional entropy. Let P_X be the empirical distribution of x^n and let Y^n be the random output of the memoryless channel when x^n is transmitted. The uncertainty (74) evaluated in x^n and Y^n becomes a random variable, which by the WLLN converges to the expectation

$$\mathbb{U}(q, X, Y) = \mathbb{E} \left[-\log_2 \frac{q(X, Y)}{\sum_{v \in \mathcal{X}} q(v, Y)} \right] \quad (75)$$

where X and Y are distributed according to $P_X p_{Y|X}$. In the expression (75), the average over n samples is replaced by the expectation according to a channel input with distribution P_X and a channel output with density $p_{Y|X}$. This makes the expression in (75) amenable for studying optimal decoding metrics q . We next consider two important cases.

B. Symbol-Metric Decoding (SMD)

For the memoryless metric

$$q_{\text{smd}}(a, b) = P_{X|Y}(a|b) \quad (76)$$

the uncertainty becomes the conditional entropy

$$\mathbb{U}(q_{\text{smd}}, X, Y) = \mathbb{E} \left[-\log_2 \frac{P_{X|Y}(X|Y)}{\sum_{v \in \mathcal{X}} P_{X|Y}(v|Y)} \right] \quad (77)$$

$$= \mathbb{H}(X|Y). \quad (78)$$

The metric (76) is actually optimal, i.e., the conditional entropy $\mathbb{H}(X|Y)$ is the smallest possible uncertainty and it is achieved by using metric $P_{X|Y}(\cdot|\cdot)$. See, e.g., [41, Sec. 5] for a derivation of the optimality.

C. Bit-Metric Decoding

In Sec. II-B, we defined the binary label χ that represents the channel input $x \in \mathcal{X}$ by m bits $\mathbf{b} = b_1 \dots b_m = \chi(x)$. A bit-metric decoder uses

$$q_{\text{bmd}}(\mathbf{b}, y) = \prod_{j=1}^m P_{B_j|Y}(b_j|y) \quad (79)$$

for which the uncertainty becomes

$$\mathbb{U}(q_{\text{bmd}}, \mathbf{B}, Y) = \sum_{j=1}^m \mathbb{H}(B_j|Y). \quad (80)$$

Practical decoders work in the log-domain and calculate for the j th bit-level the LLR

$$\ell_j(y) = \log \frac{P_{B_j|Y}(0|y)}{P_{B_j|Y}(1|y)}. \quad (81)$$

The log-domain decoding metric for the j th bit-level is then

$$q_{\log}(b_j, y) = \begin{cases} +\ell_j(y), & \text{if } b_j = 0 \\ -\ell_j(y), & \text{if } b_j = 1 \end{cases} \quad (82)$$

$$= (1 - 2b_j)\ell_j(y) = (-1)^{b_j}\ell_j(y) \quad (83)$$

and the resulting metric is

$$q_{\text{bmd}, \log}(\mathbf{b}, y) = \sum_{j=1}^m (1 - 2b_j)\ell_j(y). \quad (84)$$

Note that using

$$q(\mathbf{b}, y) = \exp[s \cdot q_{\text{bmd}, \log}(\mathbf{b}, y)], \quad s = \frac{1}{2} \quad (85)$$

recovers (79). With some manipulations, (80) becomes

$$\begin{aligned} \mathbb{U}(q_{\text{bmd}}, \mathbf{B}, \mathbf{L}) \\ = \sum_{j=1}^m \mathbb{E} \left[\log_2 \{1 + \exp[-(1 - 2B_j)L_j]\} \right] \end{aligned} \quad (86)$$

where $L_j = \ell_j(Y)$. For the equivalent metric with minimization over s , combining (72) with (84) becomes

$$\begin{aligned} \mathbb{U}^*(q_{\text{bmd}}, \mathbf{B}, \mathbf{L}) \\ = \min_{s \geq 0} \sum_{j=1}^m \mathbb{E} \left[\log_2 \{1 + \exp[-s(1 - 2B_j)L_j]\} \right]. \end{aligned} \quad (87)$$

D. Achievable Spectral Efficiencies

For memoryless channels and large n , we can now combine uncertainty and shaping set size to obtain spectral efficiency estimates. We exemplify this for symbol-metric decoding (SMD):

- 1) For sequences with distribution P_X and a memoryless channel $P_{Y|X}$, the uncertainty for large n is $\mathbb{H}(X|Y)$.
- 2) The shaping set with sequences of distribution P_X has for large n the normalized size

$$\frac{\log_2 |\mathcal{S}|}{n} = \mathbb{H}(X). \quad (88)$$

3) The achievable PS-FEC encoding rate is thus

$$\left[\frac{\log_2 |\mathcal{S}|}{n} - \mathbb{U}(q_{\text{smd}}, X, Y) \right]^+ = [\mathbb{H}(X) - \mathbb{H}(X|Y)]^+ \\ = \mathbb{I}(X; Y). \quad (89)$$

VII. OPTICAL TRANSMISSION EXPERIMENTS

A. Measurements

To fully characterize the performance of optical transceivers, transmission experiments have to be conducted. In the following, we consider the digital signal processing (DSP) and the optical channel (e.g., sampling, CD compensation, equalization, phase noise compensation) as a black box, which provides the discrete time receive samples.

For a given experimental setup (e.g., input power, transmission distance, DSP and modulation setting), we transmit a test sequence x^n , of which we measure a noisy observation y^n at the receiver. The n entries $x_1 \dots x_n$ correspond to n real-valued ASK symbols. In case of 2D modulation, two successive entries of x^n correspond to the in-phase and quadrature components of a QAM symbol; if in addition polarization division multiplexing (PDM) is applied, four successive entries of x^n correspond to the in-phase and quadrature components of the two polarizations.

B. Decoding Models

As outlined in Sec. V, an FEC decoder uses a decoding metric q to decide on the transmitted sequence. In Sec. VI, we observed that for memoryless channels with known statistics, the optimal decoding metric is an a-posteriori probability. This suggests the use of a probabilistic model and the corresponding decoding metric at the receiver.

Following [62], [63] an additive Gaussian noise model is reasonable to describe the accumulated noise at the receiver after transmission over a long, dispersion-uncompensated fiber. We therefore use the memoryless model

$$Y = \Delta X + Z \quad (90)$$

that is, the observed symbol Y is modeled as the sum of the attenuated transmitted symbol X and zero mean Gaussian noise Z with variance σ^2 . The decoder now interprets the sequences x^n and y^n as realizations of X and Y , respectively. The decoding metric for the model (90) is given by (76) as

$$q_{\text{smd}}(x, y; \sigma^2, \Delta, P_X) = P_{X|Y}(x|y; \sigma^2, \Delta, P_X) \\ \propto p_{Y|X}(y|x; \sigma^2, \Delta) P_X(x) \quad (91)$$

where

$$p_{Y|X}(y|x; \sigma^2, \Delta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \Delta x)^2}{2\sigma^2}\right). \quad (92)$$

We note that the model (90), (92) can be further refined to include non-linear effects, such as constellation point dependent attenuations and variances.

C. Model Parameter Estimation

The model parameters can be estimated either by pilots or blindly using unsupervised learning approaches, e.g., expecta-

tion maximization (EM) [64], [65]. For benchmarking practical parameter estimations, genie-aided parameter estimation using the transmitted sequence x^n and the measured sequence y^n can be done via classical maximum likelihood estimation [9, Sec. III-B] as follows:

$$\Delta = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2} \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \Delta x_i)^2 \quad (93)$$

$$P_X(v) = \frac{N(v|x^n)}{n}, \quad v \in \mathcal{X}. \quad (94)$$

Note however that in a practical system, the transmitted sequence x^n is not known at the receiver a-priori. Estimating Δ precisely is important, as it significantly influences the uncertainty. When evaluating measurements, inspecting a scatter plot of y^n is helpful to recognize a residual scaling or rotation error after the DSP.

Remark 6: The blind estimation [65] can be considered in the framework of decoding metrics as follows: first, note that blind estimation calculates the model parameters from the observation y^n , i.e., $\Delta(y^n)$, $\sigma^2(y^n)$, $P_X(y^n)$. Second, note that once the parameters are estimated, a memoryless metric is used. Together, we have

$$q(x^n, y^n) = \prod_{i=1}^n q(x_i, y_i; \Delta(y^n), \sigma^2(y^n), P_X(y^n)) \quad (95)$$

that is, blind parameter estimation is included in the general framework of decoding metrics $q(x^n, y^n)$. In particular, Theorem 2 applies and no potential rate overestimation as in the case of genie-aided estimation can occur. The blind approach only uses information that is actually available at the decoder. Having a blind estimator is often useful in optical experiments, as it helps to reveal additional assumptions and parameters which might actually not be available at the decoder. This is also helpful for shaped, rate adaptive transmission where it might be desired to automatically recognize the signaling mode, i.e., the employed distribution.

D. SMD Uncertainty

The uncertainty (62) can now be calculated as

$$\mathbb{U}^*(q_{\text{smd}}, x^n, y^n) \quad (96)$$

$$= \min_{s \geq 0} -\frac{1}{n} \sum_{i=1}^n \log_2 \frac{q_{\text{smd}}(x_i, y_i; \sigma^2, \Delta, P_X)^s}{\sum_{v \in \mathcal{X}} q_{\text{smd}}(v, y_i; \sigma^2, \Delta, P_X)^s}. \quad (97)$$

Remark 7: We note that tightening the uncertainty bound in (97) via (70) is useful as the samples x^n, y^n of the optical transmission experiment are generally not “matched” to the model (90). Employing (70) is therefore useful for real world measurements.

We evaluate the stability of the uncertainty estimate by plotting it over different lengths n and analyzing its convergence behavior. For this, the measurement x^n and y^n were taken from the transmission experiment in [9] which investigates four shaped modes OP 1 ($\mathbb{H}(X) = 5.73$ bits), OP 2 ($\mathbb{H}(X) = 5.23$ bits), OP 3 ($\mathbb{H}(X) = 4.60$ bits), OP 4 ($\mathbb{H}(X) = 4.13$ bits) based on 64-QAM. One loop consists of three spans of standard single-

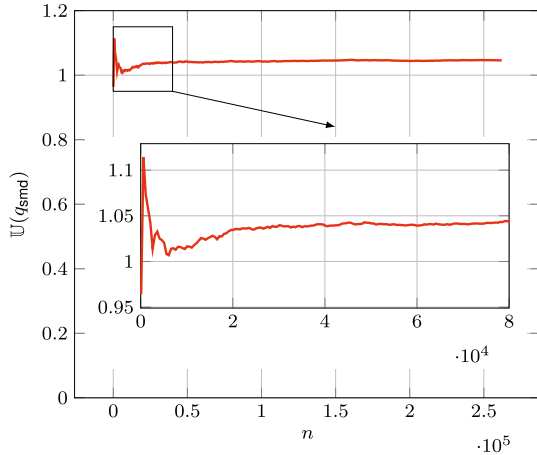


Fig. 6. Value of the SMD uncertainty estimate (97) for OP 4 while only using the first n of all collected measurement samples.

mode fiber (SSMF) with about 80 km length each. The convergence behavior of (97) is plotted in Fig. 6 for OP 4 and a transmission distance of 30 loops. For each point only the first n samples of the measurement database were used and we observe that the estimate converges after around $n = 40000$ samples.

E. Bit-Metric Decoding (BMD) Uncertainty

For binary FEC codes, we use the bit-metric decoding (BMD) metric (79). To calculate the log-domain soft information ℓ_{ij} , $i = 1, \dots, n$ and $j = 1, \dots, m$ we calculate

$$q_{\text{bmd},j}(b, y) = \sum_{x \in \mathcal{X}_j^b} q_{\text{smd}}(x, y; \sigma^2, \Delta, P_X) \quad (98)$$

$$\ell_{ij} = \log \frac{q_{\text{bmd},j}(0, y_i)}{q_{\text{bmd},j}(1, y_i)} \quad (99)$$

where $\mathcal{X}_j^b = \{x \in \mathcal{X} : \chi_j(x) = b\}$ for $j = 1, \dots, m$ and $b \in \{0, 1\}$. The uncertainty estimate is then

$$\begin{aligned} & \mathbb{U}^*(q_{\text{bmd}}, \mathbf{b}^n, \ell^n) \\ &= \min_{s \geq 0} \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m \log_2 [1 + \exp(-s(1 - 2b_{ij})\ell_{ij})]. \end{aligned} \quad (100)$$

We depict the FEC thresholds for BMD (assuming an ideal FEC code and infinite block length) in Fig. 7a. For the FEC code rate $R_{\text{fec}} = 0.8485$, the FEC threshold suggests as reach for the four modes 7.2, 11.08, 18.74 and 25.61 loops, respectively (indicated by the vertical lines in the respective OP colors).

F. Bit Error Rate (BER) Curves from Measurements

We now would like to verify the reach indicated by the FEC threshold (see Fig. 7a), by evaluating the bit error rate (BER) performance of an actual FEC code. We face the problem that the transmitted length n sequence $x_1 \dots x_n$ is usually not a codeword of the FEC code of our choice. However, we can apply a trick, called channel adapters [66] and scrambling sequences [67]. We next explain the procedure for binary codes.

For a codeword $c^{n_{\text{fec}}}$ and the binary representation $b^{n_{\text{fec}}} = \chi(x)$ of the length $n = n_{\text{fec}}/m$ transmit sequence, we calculate

the scrambling sequence $s^{n_{\text{fec}}} = s_1 \dots s_{n_{\text{fec}}}$ as

$$s_i = b_i \oplus c_i. \quad (101)$$

Using (100) we calculate the soft-information values ℓ_i , $i = 1, \dots, n_{\text{fec}}$ and pass the modified soft-information

$$\tilde{\ell}_i = \ell_i \cdot (1 - 2s_i) \quad (102)$$

to the decoder. For good enough channel conditions, the decoder should then recover $c^{n_{\text{fec}}}$ from $\tilde{\ell}^{n_{\text{fec}}}$. For linear codes, the all-zero codeword always belongs to the codebook and we can conveniently choose $c_i = 0$, $i = 1, \dots, n_{\text{fec}}$. For this all-zero codeword, (102) becomes

$$\tilde{\ell}_i = \ell_i \cdot (1 - 2b_i). \quad (103)$$

We now use the LDPC code from the upcoming IEEE 802.3ca standard [68] and try to decode one FEC frame for each of the four modes for all measured transmission distances. We hereby follow the setting of Theorem 2, which states that for a given transmit and receive pair (x^n, y^n) (where $n = n_{\text{fec}}/m$, i.e., x^n represents the modulated symbols of one FEC frame) the probability of unsuccessful decoding becomes vanishingly small when the FEC code rate R_{fec} fulfills: $R_{\text{fec}} < 1 - \mathbb{U}^*(q_{\text{bmd}})/m$. The results are displayed in Fig. 7b. As before, the vertical, solid bars indicate the respective FEC thresholds of Fig. 7a. We see that we have to backoff from the FEC thresholds such that the effective reach is slightly shorter than what the FEC threshold suggests. As pointed out in Sec. VII-E, this is because we are now using a practical, finite block length FEC code with a sub-optimal, belief propagation decoder. To show that $R_{\text{fec}}^* - R_{\text{fec}}$ remains fairly constant for all four modes, we use the respective gaps between the theoretically and practically achieved maximum transmission distances as a proxy. We note that this approach only allows to depict a general tendency rather than providing a rigorous numerical verification. For a direct comparison to $R_{\text{fec}}^* - R_{\text{fec}}$, the transmission experiment would need to be performed for different input powers with small step sizes. The slightly larger gap for OP 3 (green) is because of the loop granularity of the experiment where receive samples were only taken every second loop: The backoff would be too small for 16 loops such that successful decoding is observed only up to 14 loops.

G. Estimating Achievable Spectral Efficiencies

In Sec. VI, we derived achievable spectral efficiencies for memoryless channels. These results can be extended to more general channels, if ergodicity in the following sense holds:

- 1) The uncertainty of a specific transmit sequence x^n is the same for different transmissions.
- 2) The uncertainty is the same for different sequences from the shaping set $\mathcal{S}$.

We immediately observe that validating conditions 1) and 2) for an unknown channel requires prohibitively many measurements! On the other hand, if we observe similar uncertainty values for several transmissions of a test sequence x^n , and if different test sequences from the shaping set $\mathcal{S}$ result in similar uncertainty values, we can estimate (with some confidence) an

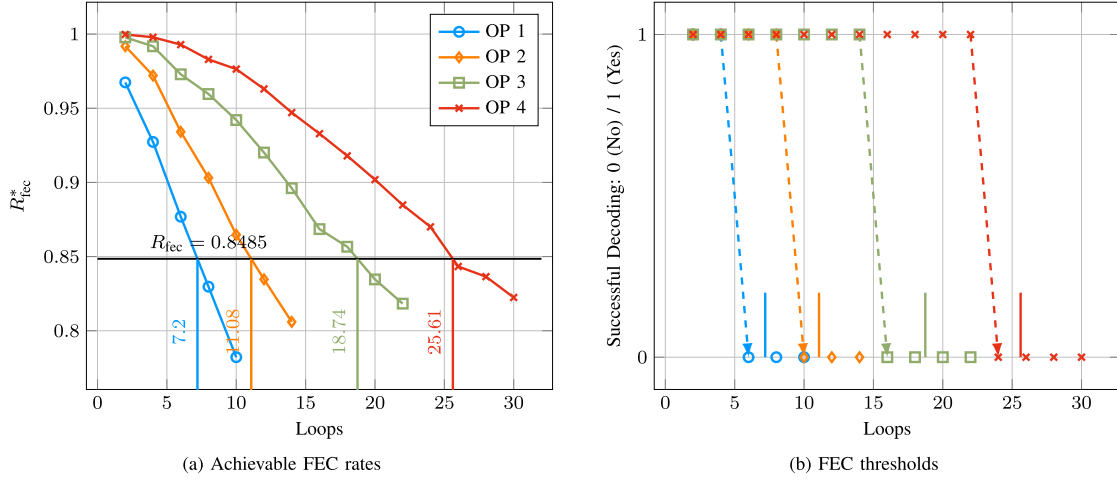


Fig. 7. In the left plot, we illustrate the achievable FEC rates for the transmission experiment of [9]. The operating points OP 1, OP 2, OP 3 and OP 4 refer to net transmission rates of 300 Gbit/s, 270 Gbit/s, 230 Gbit/s and 200 Gbit/s. In the right plot, we depict the phase transition for successful decoding for a rate $R_{\text{fec}} = 0.8485$ FEC code and compare it to the asymptotic thresholds of the left plot, which are shown by the respective solid lines. During the experiment, data was collected every second loop.

achievable spectral efficiency by

$$\left[\frac{\log_2 |\mathcal{S}|}{n} - \mathbb{U} \right]^+. \quad (104)$$

Note that the uncertainty $\mathbb{U}$ in (104) should be the largest ('worst') value measured in the conducted experiments, and not the smallest ('best').

VIII. CONCLUSIONS

In this tutorial paper, we developed a foundation for integrating PS and FEC. Topics for future research include:

- 1) The development of high throughput DM algorithms with low power consumption.
- 2) The integration of PS and FEC beyond PAS to enable, e.g., constellations with signal points at 0, PS-FEC when the PAS condition (33) is not fulfilled, and practical PS-FEC for intensity modulation, and in particular for OOK, extending previous approaches such as [69].
- 3) The performance-complexity tradeoff of higher-order and/or high-dimensional coded modulation.

APPENDIX A

PROOF OF THEOREM 1

We consider a general shaping set $\mathcal{S} \subseteq \mathcal{X}^n$ and we consider the random code ensemble that we defined in Sec. IV-A. Note that we do not condition on a specific realization of the code $\mathcal{C}$, i.e., for each index $w = 1, 2, \dots, 2^{nmR_{\text{fec}}}$, a codeword is picked uniformly at random from $\mathcal{X}^n$, independent of the other code words. For PS rate $0 \leq R_{\text{ps}} \leq 1$ we divide the FEC code into $2^{nmR_{\text{ps}}R_{\text{fec}}}$ partitions, so that the number of codewords in each partition is

$$\frac{2^{nmR_{\text{fec}}}}{2^{nmR_{\text{ps}}R_{\text{fec}}}} = 2^{nm(1-R_{\text{ps}})R_{\text{fec}}}. \quad (105)$$

The PS encoder maps message $u \in \{1, 2, \dots, 2^{nmR_{\text{ps}}R_{\text{fec}}}\}$ to a codeword in the u th partition that is in $\mathcal{S}$. In our random coding experiment, each codeword is chosen uniformly at random from $\mathcal{X}^n$ and it is not in $\mathcal{S}$ with probability

$$\frac{|\mathcal{X}|^n - |\mathcal{S}|}{|\mathcal{X}|^n} = 1 - \frac{|\mathcal{S}|}{|\mathcal{X}|^n}. \quad (106)$$

For a considered message u , the probability that of the $2^{nm(1-R_{\text{ps}})R_{\text{fec}}}$ codewords in the corresponding u th partition, none is in $\mathcal{S}$ is therefore

$$\left(1 - \frac{|\mathcal{S}|}{|\mathcal{X}|^n} \right)^{2^{nm(1-R_{\text{ps}})R_{\text{fec}}}} \leq \exp \left(- \frac{|\mathcal{S}|}{|\mathcal{X}|^n} 2^{nm(1-R_{\text{ps}})R_{\text{fec}}} \right)$$

where we used $1 - s \leq \exp(-s)$. Finally, the exponent is

$$\frac{|\mathcal{S}|}{|\mathcal{X}|^n} 2^{nm(1-R_{\text{ps}})R_{\text{fec}}} = 2^{nm \left[(1-R_{\text{ps}})R_{\text{fec}} + \frac{1}{nm} \log_2 |\mathcal{S}| \right]}. \quad (107)$$

APPENDIX B

PROOF OF THEOREM 2

Our proof is based on Markov's inequality and is similar in style to the proof of the channel coding theorem in [70].

Recall the setup from Sec. V:

- Finite constellation $\mathcal{X}$ with cardinality $|\mathcal{X}| = 2^m$.
- Random code $\mathcal{C}$ with codewords in $\mathcal{X}^n$, cardinality $|\mathcal{C}| = 2^{nmR_{\text{fec}}}$ and FEC rate $R_{\text{fec}} = \frac{\log_2 |\mathcal{C}|}{nm}$.
- The $2^{nmR_{\text{fec}}} \cdot n$ entries of the $2^{nmR_{\text{fec}}}$ codewords $C^m(w)$, $w = 1, 2, \dots, 2^{nmR_{\text{fec}}}$, are generated independently and uniformly distributed on $\mathcal{X}$.
- For the observation y^n , the decoder decides

$$\hat{W} = \underset{w \in \{1, \dots, |\mathcal{C}|\}}{\operatorname{argmax}} q(C^m(w), y^n) \quad (108)$$

where $q : \mathcal{X}^n \times \mathcal{Y}^n \rightarrow \mathbb{R}_{\geq 0}$ is a non-negative metric defined on $\mathcal{X}^n \times \mathcal{Y}^n$, and where $\mathcal{Y}$ is the possibly continuous and or/multi-dimensional output alphabet. As defined

in Section V-A, the metric is restricted to assign strictly positive scores to input-output pairs that are possible for the considered channel.

We condition on the event that index W was encoded to $C^n(W) = x^n$ and the sequence y^n was output by the channel. For notational convenience, we assume without loss of generality $W = 1$. A decoding error $\hat{W} \neq 1$ implies the existence of a $w' \neq 1$ such that

$$L(w') := \frac{q(C^n(w'), y^n)}{q(x^n, y^n)} \geq 1 \Rightarrow \sum_{w=2}^{|C|} L(w) \geq 1. \quad (109)$$

If event $\mathcal{A}$ implies event $\mathcal{B}$, then $\Pr(\mathcal{A}) \leq \Pr(\mathcal{B})$. Therefore, we have

$$\begin{aligned} \Pr(\hat{W} \neq 1 | X^n = x^n, Y^n = y^n) \\ \leq \Pr \left[\sum_{w=2}^{|C|} L(w) \geq 1 \mid X^n = x^n, Y^n = y^n \right] \end{aligned} \quad (110)$$

$$\leq \mathbb{E} \left[\sum_{w=2}^{|C|} L(w) \mid X^n = x^n, Y^n = y^n \right] \quad (111)$$

$$= q(x^n, y^n)^{-1} \mathbb{E} \left[\sum_{w=2}^{|C|} q(C^n(w), y^n) \right] \quad (112)$$

where

- Inequality in (111) follows by Markov's inequality.
- Equality in (112) follows because for $w \neq 1$, the codeword $C^n(w)$ and the transmitted codeword $C^n(1)$ were generated independently so that $C^n(w)$ and $[C^n(1), Y^n]$ are independent and the conditioning on $Y^n = y^n$ can be dropped.

The right-hand side of (112) is further bounded as

$$(112) = (|C| - 1)q(x^n, y^n)^{-1} \mathbb{E} [q(C^n, y^n)] \quad (113)$$

$$\leq |C|q(x^n, y^n)^{-1} \mathbb{E} [q(C^n, y^n)] \quad (114)$$

$$= 2^{nm R_{\text{fec}}} q(x^n, y^n)^{-1} \sum_{x^n \in \mathcal{X}^n} |\mathcal{X}|^{-n} q(x^n, y^n) \quad (115)$$

$$= 2^{-nm \left(1 + \frac{1}{nm} \log_2 \sum_{x^n \in \mathcal{X}^n} \frac{q(x^n, y^n)}{q(x^n, y^n)} - R_{\text{fec}} \right)} \quad (116)$$

where

- C^n in (113) is a random variable with entries that are independent and uniformly distributed on $\mathcal{X}$.
- Equality in (113) holds because in our random coding experiment, for each index w , we generated the codeword entries $C_1(w), C_2(w), \dots, C_n(w)$ independent and uniformly distributed on $\mathcal{X}$.

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REFERENCES

- [1] G. Böcherer, "On joint design of probabilistic shaping and forward error correction for optical systems," in *Proc. Opt. Fiber Commun. Conf. Expo.*, Mar. 2018, pp. 1–36.
- [2] G. Böcherer, F. Steiner, and P. Schulte, "Bandwidth efficient and rate-matched low-density parity-check coded modulation," *IEEE Trans. Commun.*, vol. 63, no. 12, pp. 4651–4665, Dec. 2015.
- [3] P. Schulte and G. Böcherer, "Constant composition distribution matching," *IEEE Trans. Inf. Theory*, vol. 62, no. 1, pp. 430–434, Jan. 2016.
- [4] W. G. Bliss, "Circuitry for performing error correction calculations on baseband encoded data to eliminate error propagation," *IBM Tech. Discl. Bull.*, vol. 23, pp. 4633–4634, Mar. 1981.
- [5] M. Mansuripur, "Enumerative modulation coding with arbitrary constraints and post-modulation error correction coding for data storage systems," *Proc. SPIE*, vol. 1499, Feb. 1991, pp. 72–87.
- [6] G. Böcherer, "Principles of coded modulation," Habilitation thesis, Tech. Univ. Munich, Munich, Germany, 2018. [Online]. Available: <http://www.georg-boecherer.de/boecherer2018principles.pdf>
- [7] R. A. Amjad, "Information rates and error exponents for probabilistic amplitude shaping," in *Proc. IEEE Inf. Theory Workshop*, Guangzhou, China, Nov. 2018, pp. 1–5.
- [8] F. Buchali, G. Böcherer, W. Idler, L. Schmalen, P. Schulte, and F. Steiner, "Experimental demonstration of capacity increase and rate-adaptation by probabilistically shaped 64-QAM," in *Proc. Eur. Conf. Opt. Commun.*, Valencia, Spain, 2015, Paper PDP3.4.
- [9] F. Buchali, F. Steiner, G. Böcherer, L. Schmalen, P. Schulte, and W. Idler, "Rate adaptation and reach increase by probabilistically shaped 64-QAM: An experimental demonstration," *J. Lightw. Technol.*, vol. 34, no. 7, pp. 1599–1609, Apr. 2016.
- [10] F. Buchali, W. Idler, L. Schmalen, G. Böcherer, P. Schulte, and F. Steiner, "Probabilistically shaped QAM for independent reach, spectral efficiency and bit-rate adaptation," in *Proc. Eur. Conf. Opt. Commun.*, Düsseldorf, Germany, Sep. 2016, Paper W.1.C.
- [11] J. Cho *et al.*, "Trans-Atlantic field trial using probabilistically shaped 64-QAM at high spectral efficiencies and single-carrier real-time 250-Gb/s 16-QAM," in *Proc. Opt. Fiber Commun. Conf.*, 2017, Paper Th.5.B.3.
- [12] A. Ghazisaeidi *et al.*, "Advanced C+L-band transoceanic transmission systems based on probabilistically shaped PDM-64QAM," *J. Lightw. Technol.*, vol. 35, no. 7, pp. 1291–1299, Apr. 2017.
- [13] W. Idler *et al.*, "Field trial of a 1 Tb/s super-channel network using probabilistically shaped constellations," *J. Lightw. Technol.*, vol. 35, no. 8, pp. 1399–1406, Apr. 2017.
- [14] J. Li *et al.*, "Field trial of probabilistic-shaping-programmable real-time 200-Gb/s coherent transceivers in an intelligent core optical network," in *Proc. Asia Commun. Photon. Conf.*, 2018, pp. 1–3.
- [15] M. P. Yankov *et al.*, "Constellation shaping for WDM systems using 256QAM/1024QAM with probabilistic optimization," *J. Lightw. Technol.*, vol. 34, no. 22, pp. 5146–5156, Nov. 2016.
- [16] R. Maher, K. Croussore, M. Laueremann, R. Going, X. Xu, and J. Rahn, "Constellation shaped 66 GbD DP-1024QAM transceiver with 400 km transmission over standard SMF," in *Proc. Eur. Conf. Opt. Commun.*, 2017, Paper PDPB.2.
- [17] Z. Qu, S. Zhang, and I. B. Djordjevic, "Universal hybrid probabilistic-geometric shaping based on two-dimensional distribution matchers," in *Proc. Opt. Fiber Commun. Conf.*, San Diego, CA, USA, Mar. 2018, Paper M4E.4.
- [18] S. L. I. Olsson, J. Cho, S. Chandrasekhar, X. Chen, E. C. Burrows, and P. J. Winzer, "Record-high 17.3-bit/s/Hz spectral efficiency transmission over 50 km using probabilistically shaped PDM 4096-QAM," in *Proc. Opt. Fiber Commun. Conf.*, Mar. 2018, Paper Th.4.C.5.
- [19] F. Steiner *et al.*, "Experimental verification of rate flexibility and probabilistic shaping by 4D signaling," in *Proc. Opt. Fiber Commun. Conf.*, Mar. 2018, Paper M.4.E.3.
- [20] O. Geller, R. Dar, M. Feder, and M. Shtaiif, "A shaping algorithm for mitigating inter-channel nonlinear phase-noise in nonlinear fiber systems," *J. Lightw. Technol.*, vol. 34, no. 16, pp. 3884–3889, Aug. 2016.
- [21] M. P. Yankov, K. J. Larsen, and S. Forchhammer, "Temporal probabilistic shaping for mitigation of nonlinearities in optical fiber systems," *J. Lightw. Technol.*, vol. 35, no. 10, pp. 1803–1810, May 2017.

- [22] J. Renner *et al.*, "Experimental comparison of probabilistic shaping methods for unrepeated fiber transmission," *J. Lightw. Technol.*, vol. 35, no. 22, pp. 4871–4879, Nov. 2017.
- [23] E. Sillekens *et al.*, "A simple nonlinearity-tailored probabilistic shaping distribution for square QAM," in *Proc. Opt. Fiber Commun. Conf.*, Mar. 2018, Paper M.3.C.4.
- [24] S. Chandrasekhar *et al.*, "High-spectral-efficiency transmission of PDM 256-QAM with parallel probabilistic shaping at record rate-reach trade-offs," in *Proc. Eur. Conf. Opt. Commun.*, 2016, Paper Th.3.C.1.
- [25] J. Cho, S. Chandrasekhar, R. Dar, and P. J. Winzer, "Low-complexity shaping for enhanced nonlinearity tolerance," in *Proc. Eur. Conf. Opt. Commun.*, Sep. 2016, pp. 467–469.
- [26] P. Schulte, F. Steiner, and G. Böcherer, "Four dimensional probabilistic shaping for fiber-optic communication," in *Proc. Signal Process. Photonic Commun.*, Jul. 2017, Paper SpM2F.5.
- [27] G. Böcherer, P. Schulte, and F. Steiner, "High throughput probabilistic shaping with product distribution matching," arXiv:1702.07510, 2017. [Online]. Available: <http://arxiv.org/abs/1702.07510>
- [28] G. Böcherer, F. Steiner, and P. Schulte, "Fast probabilistic shaping implementation for long-haul fiber-optic communication systems," in *Proc. Eur. Conf. Opt. Commun.*, Gothenburg, Sweden, Sep. 2017, Paper Tu.2.D.3.
- [29] Y. Gültekin, F. Willems, W. van Houtum, and S. Serbetli, "Approximate enumerative sphere shaping," in *Proc. IEEE Int. Symp. Inf. Theory*, Jun. 2018, pp. 676–680.
- [30] F. Steiner, G. Böcherer, and P. Schulte, "Approaching waterfilling capacity of parallel channels by higher order modulation and probabilistic amplitude shaping," in *Proc. 52nd Annu. Conf. Inf. Sci. Syst.*, Mar. 2018, pp. 1–6.
- [31] T. Yoshida, M. Karlsson, and E. Agrell, "Hierarchical distribution matching for probabilistically shaped coded modulation," *J. Lightw. Technol.*, doi: [10.1109/JLT.2019.2895065](https://doi.org/10.1109/JLT.2019.2895065).
- [32] J. Cho, "Prefix-Free code distribution matching for probabilistic constellation shaping," Oct. 2018, arXiv:1810.02411.
- [33] T. Fehenberger, D. S. Millar, T. Koike-Akino, K. Kojima, and K. Parsons, "Multiset-partition distribution matching," *IEEE Trans. Commun.*, doi: [10.1109/TCOMM.2018.2881091](https://doi.org/10.1109/TCOMM.2018.2881091).
- [34] D. Millar, T. Fehenberger, T. Koike-Akino, K. Kojima, and K. Parsons, "Distribution matching for high spectral efficiency optical communication with multiset partitions," *J. Lightw. Technol.*, to be published.
- [35] P. Schulte and F. Steiner, "Divergence-optimal fixed-to-fixed length distribution matching with shell mapping," *IEEE Wireless Commun. Lett.*, doi: [10.1109/LWC.2018.2890595](https://doi.org/10.1109/LWC.2018.2890595).
- [36] F. Steiner, G. Böcherer, and G. Liva, "Protograph-based LDPC code design for shaped bit-metric decoding," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 2, pp. 397–407, Feb. 2016.
- [37] F. Steiner and P. Schulte, "Design of robust, protograph based LDPC codes for rate-adaptation via probabilistic shaping," in *Proc. 9th Int. Symp. Turbo Codes Iterative Inf. Process.*, Sep. 2016, pp. 56–60.
- [38] C. Pan and F. R. Kschischang, "Probabilistic 16-QAM shaping in WDM systems," *J. Lightw. Technol.*, vol. 34, no. 18, pp. 4285–4292, Sep. 2016.
- [39] L. Schmalen, "Probabilistic constellation shaping: Challenges and opportunities for forward error correction," in *Proc. Opt. Fiber Commun. Conf.*, Mar. 2018, Paper M.3.C.1.
- [40] A. Sheikh, A. Graell i Amat, G. Liva, and F. Steiner, "Probabilistic amplitude shaping with hard decision decoding and staircase codes," *J. Lightw. Technol.*, vol. 36, no. 9, pp. 1689–1697, May 2018.
- [41] G. Böcherer, "Achievable rates for shaped bit-metric decoding," arXiv preprint, 2016. [Online]. Available: <http://arxiv.org/abs/1410.8075>
- [42] G. Böcherer, "Achievable rates for probabilistic shaping," arXiv preprint, 2017. [Online]. Available: <https://arxiv.org/abs/1707.01134>
- [43] G. Kaplan and S. Shamai (Shitz), "Information rates and error exponents of compound channels with application to antipodal signaling in a fading environment," *AEÜ*, vol. 47, no. 4, pp. 228–239, 1993.
- [44] A. Alvarado, E. Agrell, D. Lavery, R. Maher, and P. Bayvel, "Replacing the soft-decision FEC limit paradigm in the design of optical communication systems," *J. Lightw. Technol.*, vol. 33, no. 20, pp. 4338–4352, Oct. 2015.
- [45] J. Cho, L. Schmalen, and P. J. Winzer, "Normalized generalized mutual information as a forward error correction threshold for probabilistically shaped QAM," in *Proc. Eur. Conf. Opt. Commun.*, Sep. 2017, Paper M.2.D.2.
- [46] T. Yoshida, M. Karlsson, and E. Agrell, "Performance metrics for systems with soft-decision FEC and probabilistic shaping," *IEEE Photon. Technol. Lett.*, vol. 29, no. 23, pp. 2111–2114, Dec. 2017.
- [47] F. Steiner and G. Böcherer, "Comparison of geometric and probabilistic shaping with application to ATSC 3.0," in *Proc. 11th Int. ITG Conf. Syst. Commun. Coding*, Hamburg, Germany, Feb. 2017, pp. 1–6c.
- [48] D.-S. Shiu and J. M. Kahn, "Shaping and nonequiprobable signaling for intensity-modulated signals," *IEEE Trans. Inf. Theory*, vol. 45, no. 7, pp. 2661–2668, Nov. 1999.
- [49] P. J. Winzer and R.-J. Essiambre, "Advanced optical modulation formats," *Proc. IEEE*, vol. 94, no. 5, pp. 952–985, May 2006.
- [50] T. A. Eriksson, M. Chagnon, F. Buchali, K. Schuh, S. ten Brink, and L. Schmalen, "56 Gbaud probabilistically shaped PAM8 for data center interconnects," in *Proc. Eur. Conf. Opt. Commun.*, 2017, pp. 1–6.
- [51] A. K. Khandani and P. Kabal, "Shaping multidimensional signal spaces. I. Optimum shaping, shell mapping," *IEEE Trans. Inf. Theory*, vol. 39, no. 6, pp. 1799–1808, Nov. 1993.
- [52] T. Cover, "Enumerative source encoding," *IEEE Trans. Inf. Theory*, vol. 19, no. 1, pp. 73–77, Jan. 1973.
- [53] J. Forney, G. R. Gallager, G. Lang, F. Longstaff, and S. Qureshi, "Efficient modulation for band-limited channels," *IEEE J. Sel. Areas Commun.*, vol. 2, no. 5, pp. 632–647, Sep. 1984.
- [54] F. R. Kschischang and S. Pasupathy, "Optimal nonuniform signaling for Gaussian channels," *IEEE Trans. Inf. Theory*, vol. 39, no. 3, pp. 913–929, May 1993.
- [55] G. Böcherer and R. Mathar, "Matching dyadic distributions to channels," in *Proc. Data Compression Conf.*, 2011, pp. 23–32.
- [56] G. Böcherer and B. C. Geiger, "Optimal quantization for distribution synthesis," *IEEE Trans. Inf. Theory*, vol. 62, no. 11, pp. 6162–6172, Nov. 2016.
- [57] R. Johannesson and K. S. Zigangirov, *Fundamentals of Convolutional Coding*, 2nd ed. Hoboken, NJ, USA: Wiley, 2015.
- [58] S.-Y. Chung, G. D. Forney, T. J. Richardson, and R. Urbanke, "On the design of low-density parity-check codes within 0.0045 dB of the Shannon limit," *IEEE Commun. Lett.*, vol. 5, no. 2, pp. 58–60, Feb. 2001.
- [59] D. Divsalar, S. Dolinar, C. R. Jones, and K. Andrews, "Capacity-approaching protograph codes," *IEEE J. Sel. Areas Commun.*, vol. 27, no. 6, pp. 876–888, Aug. 2009.
- [60] L. Schmalen, A. Alvarado, and R. Rios-Müller, "Performance prediction of nonbinary forward error correction in optical transmission experiments," *J. Lightw. Technol.*, vol. 35, no. 4, pp. 1015–1027, Feb. 2017.
- [61] C. Häger, A. Graell i Amat, F. Brännström, A. Alvarado, and E. Agrell, "Improving soft FEC performance for higher-order modulations via optimized bit channel mappings," *Opt. Express*, vol. 22, no. 12, pp. 14, 544–14,558, Jun. 2014.
- [62] R. Dar, M. Feder, A. Mecozzi, and M. Shtaif, "Properties of nonlinear noise in long, dispersion-uncompensated fiber links," *Opt. Express*, vol. 21, no. 22, pp. 25 685–25 699, Nov. 2013.
- [63] P. Poggiolini, G. Bosco, A. Carena, V. Curri, Y. Jiang, and F. Forghieri, "The GN-model of fiber non-linear propagation and its applications," *J. Lightw. Technol.*, vol. 32, no. 4, pp. 694–721, Feb. 2014.
- [64] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *J. Roy. Statist. Soc.*, vol. 39, no. 1, pp. 1–38, 1977.
- [65] F. Steiner, P. Schulte, and G. Böcherer, "Blind decoding-metric estimation for probabilistic shaping via expectation maximization," in *Proc. Eur. Conf. Opt. Commun.*, Sep. 2018, Paper Th.1.H.3.
- [66] J. Hou, P. H. Siegel, L. B. Milstein, and H. D. Pfister, "Capacity-approaching bandwidth-efficient coded modulation schemes based on low-density parity-check codes," *IEEE Trans. Inf. Theory*, vol. 49, no. 9, pp. 2141–2155, Sep. 2003.
- [67] L. Schmalen, F. Buchali, A. Leven, and S. ten Brink, "A generic tool for assessing the soft-FEC performance in optical transmission experiments," *IEEE Photon. Technol. Lett.*, vol. 24, no. 1, pp. 40–42, Jan. 2012.
- [68] IEEE LAN/MAN Standards Committee, "Draft standard for ethernet amendment: Physical layer specifications and management parameters for 25 Gb/s, 50 Gb/s, and 100 Gb/s passive optical networks," Tech. Rep. IEEE P802.3ca/D1.0, Mar. 2018.
- [69] A. Git, B. Matuz, and F. Steiner, "Protograph-based LDPC code design for probabilistic shaping with on-off keying," in *Proc. Ann. Conf. Inf. Sci. Syst.*, Mar. 2019, pp. 1–6.
- [70] Y. Lomnitz and M. Feder, "A simpler derivation of the coding theorem," arXiv preprint, May 2012. [Online]. Available: <http://arxiv.org/1205.1389>

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